NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND

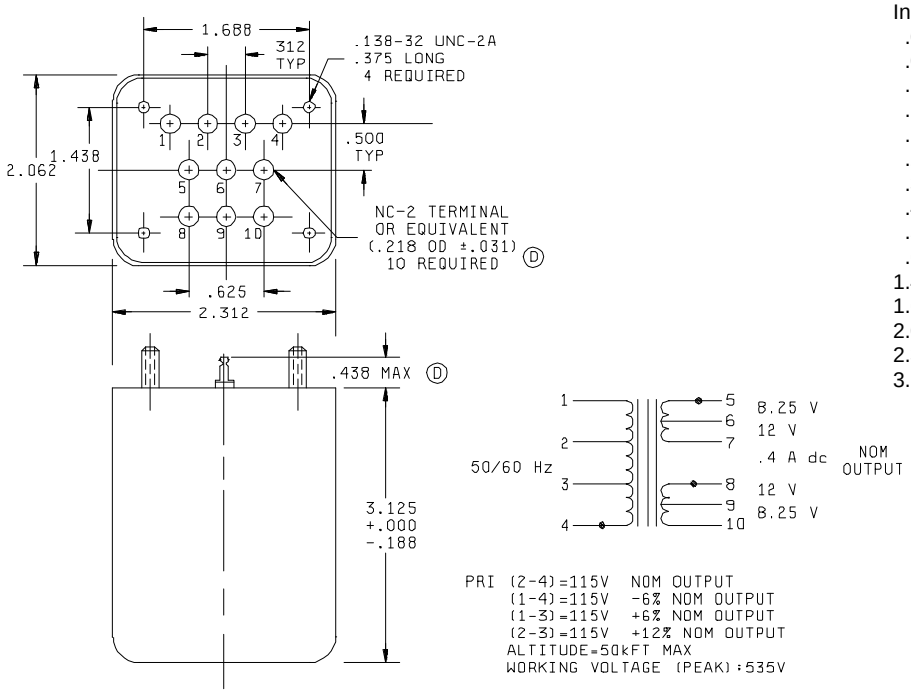
MIL-PRF-27/157D
7 March 1995
SUPERSEDING
MIL-T-27/157C
23 July 1990

PERFORMANCE SPECIFICATION SHEET

TRANSFORMER, POWER, STEPDOWN, TF4S03FA

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-T-27.



Inches	mm
.016	0.41
.031	0.79
.125	3.18
.138	3.51
.188	4.78
.218	5.54
.375	9.52
.438	11.13
.500	12.70
.625	15.88
1.438	36.53
1.688	42.88
2.062	52.37
2.312	58.72
3.125	79.38

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance on overall case dimensions is +.000, -.125 inch.
4. Tolerance on terminal position dimensions is ±.016 inch.
5. Marking shall be on the side of the case.

FIGURE 1. Dimensions and configuration.

Ⓓ denotes changes

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REQUIREMENTS: (When numbers in parentheses, i.e., (1-2) are used, they indicate the winding and the extreme terminals of the windings.)

Electrical ratings:

TABLE I. Electrical ratings.

Dash no.	Primary voltage	Secondary voltage			
		(5-6)	(6-7)	(8-9)	(9-10)
01	(2-4): 115 V, 50/60 Hz	8.25 V	12.0 V	12.0 V	8.25 V
	(1-4): 115 V, 50/60 Hz	7.76 V	11.28 V	11.28 V	7.76 V
	(1-3): 115 V, 50/60 Hz	8.74 V	12.72 V	12.72 V	8.74 V
	(2-3): 115 V, 50/60 Hz	9.24 V	13.44 V	13.44 V	9.24 V

Primary current (1-4): 0.15 ampere.

Secondary current:

(5-7): 0.4 ampere.

(8-10). 0.4 ampere.

VA: 16.

Working voltage (peak): 535 volts

Design and construction:

Dimensions and configuration: See figure 1.

Duty cycle: Continuous.

Case. Metal encased.

(D) Terminals: Solder terminal Lundey #199E or equivalent.

(D) Terminal height: .438 inch (11.13 mm) maximum.

(D) Weight: 32.63 ounce maximum.

Altitude: 50,000 feet maximum.

Operating temperature range: -55°C to +130°C.

Terminal strength: MIL-STD-202, method 211, test condition A, 5.0 pounds and test condition E, 8.0 ounce-inches.

Shock: MIL-STD-202, method 213, test condition H.

Dielectric withstanding voltage:

At sea level: 1,500 volts rms.

At reduced barometric pressure 670 volts rms.

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Electrical characteristics:

No load: With 115 V and 60 Hz in (2-4):

Current in (2-4): 120 mA maximum.

Power in (2-4): 1.5 watts maximum.

Voltage across (5-7): 22.7 volts, ±3 percent.

Voltage across (8-10): 22.7 volts, ±3 percent.

Rated load: With 115 volts and 60 Hz across (2-4):

Voltage across (5-7): 20.25 volts, ±5 percent.

Voltage across (8-10): 20.25 volts, ±5 percent.

DC resistance:

(1-4): 27.4 ohms, ±25 percent

(8-10): 2.6 ohms, ±25 percent.

(5-7): 2.43 ohms, ±25 percent.

Polarity: Additive, with terminals 1 and 5, and 7 and 8 connected.

Temperature rise: 40°C with 115 volts rms, 60 Hz across (2-4) at an ambient temperature of 90°C

Marking location: See figure 1.

(D) Part or Identifying Number (PIN): M27/157-01.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, ME, MI
Navy - AS, MC, OS
Air Force - 17, 19, 99
DLA - ES

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5950-0842)

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